

ABSTRACT

Title: THE EFFECTIVENESS OF ALGEBRAZZLE QUEST IN STRENGTHENING THE COMPETENCE OF GRADE 7 STUDENTS IN SOLVING ALGEBRAIC EQUATIONS AT D.Q. LIWAG NATIONAL HIGH SCHOOL

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This study determined the effectiveness of Algebrazzle Quest, a puzzle game-based Strategic Intervention Material (SIM), in strengthening the competence of Grade 7 students in solving algebraic equations at D.Q. Liwag National High School during the school year 2025–2026. Specifically, it aimed to identify the competency level of students before and after using Algebrazzle Quest, determine the effect size of the intervention, and assess the significant difference in students' competency levels before and after its utilization. The researchers employed a quantitative research design using a one-group pre-test and post-test approach. The respondents were 28 Grade 7 students from a less-performing section selected purposely. Data were analyzed using mean, mean percentage score, standard deviation, paired sample t-test, and Cohen's *d*. Findings revealed that students initially had a low level of competency with a pre-test mean percentage score of 44.52%. After using Algebrazzle Quest, their performance improved to a post-test mean percentage score of 65.60%, interpreted as a moderate level of competency. The computed paired sample t-test showed a significant difference between the pre-test and post-test scores, with a mean difference of -6.32143, t-value of -35.400, and p-value of $< .001$. The computed effect size using Cohen's *d* was -6.690, interpreted as a very large effect. These results indicate that Algebrazzle Quest had a significant and meaningful impact in strengthening students' competence in solving algebraic equations. The study concluded that Algebrazzle Quest effectively improves Grade 7 students' mathematical competence and promotes a more engaging, interactive, and student-centered learning experience. The study recommends the integration of similar Strategic Intervention Materials in mathematics instruction to help improve students' performance, confidence, and participation. Future researchers may also explore the use of game-based SIMs in other mathematical topics and grade levels.

Keywords: *Algebrazzle Quest, Strategic Intervention Material, Algebraic Equations, Game-Based Learning, Mathematical Competence*

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